

DHANUSHKA WEERAKOON

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EDUCATION

B.Sc. (Hons.) in Mechanical Engineering

2022 - Present

University of Moratuwa, Sri Lanka

Current GPA: 3.312

Bachelor of Information Technology (Level 1 Undergraduate)

2022 - Present

University of Moratuwa, Sri Lanka (External)

Current GPA: 3.58

G.C.E. Advanced Level Examination

2020

Maliyadeva College, Kurunegala, Sri Lanka

Z-score: 2.1619 (3A passes)

PROJECTS

Design and Development of a Hydrodynamic Tunnel for Experimental Fluid Dynamics

- Designed and developed a hydrodynamic tunnel to achieve controlled laminar flow for CFD validation. Focused on flow conditioning, CAD design, and CFD analysis to ensure accurate flow behavior.

CFD Analysis of Compressible Flow in Convergent-Divergent Nozzle

- Performed 2D CFD simulations using ANSYS Fluent to analyze compressible flow and shock formations. Evaluated pressure, temperature, velocity, and density variations under different NPR conditions.

White Pepper Processing Machine

- Designed a motorized double-roller mechanism to automate white pepper production. Improved efficiency, reduced labor, and enhanced product quality compared to traditional manual methods.

Automatic Feeding System for Threaded Rods

- Developed an automated feeding mechanism for short-length rods to eliminate manual handling. Improved production efficiency while reducing labor costs and safety risks.

Agricultural Digging Mechanism – 4-Bar Linkage System

- Designed a crank-rocker linkage to perform digging and refilling in a single motion. Modeled and simulated in MATLAB and validated using a physical prototype.

Automatic Solar Drying Robot

- Developed an autonomous robot using Arduino and multiple sensors to track sunlight and avoid obstacles. Integrated GPS and rain detection for real-time environmental response.

Gearbox Design for Fine Material Washer

- Designed a complete gearbox system with ISO-standard calculations for gears, shafts, and housing. Created full CAD models for industrial applications in recycling systems.

EV Battery Pack Design & Thermal Optimization

- Worked on battery pack optimization including cold plate design and thermal simulations. Improved thermal performance and structural integrity for EV applications.

Heat Sink Design & Thermal Analysis

- Analyzed forced convection heat transfer using analytical and CFD methods. Optimized thermal performance using Nusselt and Reynolds number evaluations.

Design & Fabrication of Painting Fume Extraction System

- Designed a ventilation and filtration system to reduce VOC emissions and particulate exposure in industrial spray painting operations. Integrated airflow design, ducting, and filtration to improve worker safety and ensure environmental compliance.

Automated Smart Medibox

- Developed an IoT-enabled smart medical box using Arduino and Node-RED to monitor environmental conditions and automate medication dispensing. Integrated sensors, MQTT-based communication, and time-based control (NTP) for remote monitoring and operation.

Pedestrian Overpass Structural Design

- Performed structural analysis considering dead, live, wind, and impact loads. Developed a design compliant with industry structural standards.

Laser Welding Process Optimization – EV Battery Systems

- Conducted experimental testing to optimize laser welding parameters for battery pack assembly. Improved weld quality and joint strength for high-current applications.

EXPERIENCE

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- Mechanical Engineer Intern – VEGA Innovations (Pvt) Ltd** 2024 – 2025
- Contributed to the design and development of Li-ion battery packs for electric vehicles.
 - Worked on laser welding processes, including parameter optimization for cylindrical cell welding.
 - Designed and analyzed heat sinks using SolidWorks and Abaqus FEA under thermal and vibration conditions.
 - Developed a pneumatic non-destructive testing system for EV battery cooling systems with a focus on long-term durability.
 - Strengthened skills in CAD/CAE, thermal management, structural analysis, and experimental design.
- BYD EV Demonstrator – John Keells Holdings PLC (EV Motor Show)** 2024
- Demonstrated EV technologies including batteries, powertrains, and advanced vehicle systems during a 3-day exhibition.
 - Effectively communicated technical concepts to diverse audiences, improving technical presentation skills.
- Trainee Cashier – People’s Bank** 2020 – 2022
- Handled cash flow operations and customer transactions in a high-volume environment.
 - Developed strong communication, teamwork, and stress management skills.
- Web Content Developer – Webdelico (Pvt) Ltd** 2020 – 2021
- Worked on WordPress-based web development, database management, and content optimization in a shopping website.
 - Gained experience in coding in Python, Java, C++, and JavaScript while collaborating in a team environment.

SKILLS

Languages: English (professional proficiency), Sinhala (native proficiency)

Software:

- **3D Modeling:** SolidWorks, Solid Edge, AutoCAD
- **Simulation:** ANSYS, Abaqus, MATLAB, Tecnomatix, xfoil
- **Control & Automation:** Siemens STEP7 MicroWIN, Yokogawa WideField
- **Video Editing:** DaVinci Resolve, CapCut
- **PCB Design:** EasyEDA, Wokwi
- **Design & Visualization:** Adobe Photoshop, Canva

Programming: Python, C++, JavaScript, Java, C#, SQL

Soft Skills: Leadership, Communication, Public Speaking, Teamwork, Project Management

VOLUNTEERING

Chairman – IET On Campus, University of Moratuwa 2025 – Present

Senior Member – IESL Student Chapter, University of Moratuwa 2025 – Present

Graphic Pillar Member – Mora Esports Community 2023 – 2025

Committee Member (Exhibit) – EXMO 2023, University of Moratuwa 2023

Member – Mora Hiking Club 2022 – Present

CERTIFICATIONS & AWARDS

Certifications:

- MATLAB Onramp – MathWorks 2023
- ReactJS Basics – Meta (Coursera) 2023
- Python for Beginners – University of Moratuwa (CODL) 2023
- Web Designing for Beginners – University of Moratuwa (CODL) 2023
- Advanced Java Programming – MIHA Institute 2022
- Python Advanced – MIHA Institute 2021

Awards:

- IMechE SL Group Awards – Final Year Project Award 2025
- President Scout Award 2019
- School Special Gold Medal – Best Performing Senior Scout 2019
- School Colors Award 2019
- Cub Scout Gold Star Award 2011

REFERENCES

Dr. Lihil U. Subasinghe

PhD (SG), BSc Eng Hons (SL), AMIMechE, AMIE(SL)

Senior Lecturer,

Department of Mechanical Engineering,

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